

Neriah Okolo

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EDUCATION

Rice University — Houston, TX

Aug 2025 - Expected May 2029

B.S. Computer Science (AI Concentration), Minor in Business & French

TECHNICAL SKILLS

Languages: Python, React.JS, Node.JS, Next.JS, ReactNative, Typescript, JavaScript, Tailwind, HTML, CSS.

Tools: GitHub, Git, VS Code, Microsoft Suite, SimplePlot, Basic Data Analysis, Codeskulptor

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Algorithmic Thinking, Applied Computational Mathematics, Computational thinking, Entrepreneurship

ACHIEVEMENTS

National Merit Scholar (2023); International Astronomy & Astrophysics Competition (**IAAC**)— Bronze Medal (2024); National Society of Black Engineers **APEX** Honors; **NTIC** National Mathematics Competition Olympiad (**NMCO**)— Bronze Medal; Kangourou Sans Frontières (**KSF**) Mathematics Competition — Bronze Medal (2022); **QuestBridge Scholar** (2024) - part of 8% nation-wide

EXPERIENCE

Handshake (Project Hedgehog)

Houston, TX

AI Research & Vision-Language Model Contractor

Oct 2025 – Dec 2025

- Developed multimodal image-editing AI to edit complex videos and images across different formats.
- Annotated 100+ sequential video frame examples to train a model resulting in high-accuracy and performance.
- Engineered precise edit instructions to improve model consistency and edit controllability.
- Increased model edit-accuracy by 25% across internal benchmarks through structured visual-delta documentation and iterative prompt refinement.
- Selected for invite-only AI research cohort supporting next-generation VLM-based image editing systems.

CodeWorld Innovations

Abuja, Nigeria

Software & Web Developer Intern

June 2021 - Aug 2023

- Designed and launched *CartMax*, an e-commerce web-application using HTML, CSS, and JavaScript, to reduce the hassle of in-person store visit, making it easier for customers to find their desired items.
- Reached 500 active users, reducing average search time by 40%.
- Leveraged social media sentiment research to create data-driven campaigns, boosting engagement by 65% and adding 1,200+ followers.

PROJECTS

L-mail - School Maintenance Workflow Platform

2023 – 2024

- Designed and deployed a web-based job-order ticketing system to replace slow, paper-based campus maintenance workflows.
- Reduced campus-wide paper usage by > 20%, saving over \$1,600 monthly through digital request routing and automated resolution tracking.
- Hosted the platform on a local school server, enabling real-time maintenance updates for students, faculty, and staff.
- Implemented automatic clearing of resolved tickets, improving storage efficiency and long-term system reliability for 1,000+ users.

Stock Market Model

2025

- Implemented a Markov Chain model in Python to predict stock-price movements for Google, First Solar Inc., and the Dow Jones Industrial Average using two years of historical data.
- Achieved 78% directional accuracy across test periods and evaluated model performance using transition-probability matrices.
- Designed modular Python package with unit tests, clear separation of modeling, prediction, and evaluation logic.

Six Degrees Graph Traversal System

2025

- Built a Python-based graph system connecting 5,000 actors across 3,400 movies; implemented BFS traversal to compute shortest path queries under 0.03s, demonstrating six-degrees-of-separation phenomena with $O(V+E)$ efficiency.

LEADERSHIP

Treasurer, Rice African Students Association

2025 – Present

Managed \$10,000+ organizational budget and led fundraising initiatives raising \$6,000+.

Undergraduate Liaison, Rice Black Men's Association

2025 – Present

Served as the primary representative for first-year students, coordinating onboarding efforts and relaying cohort feedback to executive leadership.

ORGANIZATIONS

Colorstack, National Society of Black Engineers (**NSBE**), RiceApps, Rice African Students Association (**RASA**), Black Students Association (**BSA**)